

Ethan Stace

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Education

University of Florida <i>BS in Physics, Mathematics, Minor in Electrical Engineering (GPA: 3.93/4.00)</i> • Relevant Coursework: Solid State Physics, Solid State Devices, Statistical Mechanics, Quantum Mechanics I & II, Circuits, Machine Learning for Physics, Electromagnetism I & II	August 2023 - May 2027 Gainesville, FL
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Experience

Undergraduate Researcher in Electrical Engineering <i>University of Florida</i> • Advisor: Prof. Philip Feng (Electrical & Computer Engineering) • Developing oscillator machine to perform multiply-accumulate operations in parallel • Developing code on an FPGA to perform real-time digital signal processing of oscillator machine data • Characterizing resonance/synchronization properties of systems of coupled MEMS oscillators	November 2025 - Present Gainesville, FL
UT-Dallas Physics REU Intern <i>The University of Texas at Dallas</i> • Advisor: Prof. Aaron Smith (Astrophysics) • Developed the Atom module in Cosmic Lyman-α Transfer Code, a new module for simulating multi-lined, coupled resonant scattering in astrophysical media • Developed novel analytical solution to radiative transfer equation to model multi-lined transfer networks • Used Atom module to study Lyman-α and Deuterium-injected systems	May 2025 - August 2025 Dallas, TX
Undergraduate Researcher in Plasma Astrophysics <i>University of Florida</i> • Advisor: Prof. Siyao Xu (Physics) • Analyzed data from HPC simulations to study physics of weakly-collisional positron-electron plasma • Developed novel algorithm to measure mean free path by tracking sharp changes in trajectories, and how the mean free path changed at different length and energy regimes • Showed that mean free path distribution is non-Maxwellian, matching expected behavior from anisotropy of weakly-collisional plasmas.	October 2024 - May 2025 Gainesville, FL

Extracurricular Activities

Society of Physics Students Outreach Chair, (2x) • Lead the outreach efforts of the Society of Physics Students, both within the department and with outside organizations • Created the outreach committee to brainstorm ideas and organize events • Lead the Mentor-Mentee program, responsible for speed-dating, setting up social events, and ensuring healthy communication between matches	Fall 2024 - May 2026
SCEECs Summer School on Extreme Electrodynamics and Plasma Physics Student • Attended intensive two weeks of graduate-level lectures on plasma astrophysics at The University of Texas at San Antonio taught by renowned scientists from academic institutions across the world	June 2025
SIMIODE Challenge Using Differential Equations Modeling Competitor • Joined a three-person team to compete in an international math-modeling competition • Created infectious disease model using Lotka-Volterra equations to model the spread of violence through human populations, then analyzed the behavior of solutions using numerical methods • Scored the highest raw score (4.857/5.000) out of all 124 teams, earned <i>Outstanding Team Award</i>	November 2023

Publications, Posters, and Presentations

Publications

- **Stace, E.**, Smith, A., Lorinc, K., Nebrin, O. "Multi-Resonant-Line Radiative Transfer: Lyman- α Fine Structure and Deuterium Coupling." Publications of the Astronomical Society of Australia. <https://arxiv.org/abs/2511.20580>. (Accepted 1/22/26)

Conference Papers

- Caplice, T., Kaiser, T., **Stace, E.**, Rouke, M., Casella, C., Shaw, S., Feng, X. L. "Experimental Demonstration of Nanomechanical Oscillator Ising Machine for Solving Optimization Problems." Hilton Head Workshop 2026.

Talks

- **Stace, E.**, Smith, A., Lorinc, K., Nebrin, O. "Multi-Resonant-Line Radiative Transfer: Lyman- α Fine Structure and Deuterium Coupling." 2026 APS Global Physics Summit.

Posters

- **Stace, E.**, Smith, A., Lorinc, K. "Leveling Up: Resonant Scattering within Multi-Level Atoms." 2025 Summer Platform for Undergraduate Research (SPUR) Symposium. July 31, 2025
- **Stace, E.**, Xu, Siyao. "Numerical Measurement of Mean Free Path of High Energy Weakly Plasma." 2025 Annual Undergraduate Research Poster Session. April 25, 2025

Awards

- *CLAS University Scholars Program*: Funded by University of Florida for undergraduate research during 2025-2026 academic year
- *DAP Travel Grant*: Travel reimbursement towards 2026 APS Global Physics Summit
- *Garrett Endowment for Physics Education* (4x)
- *2025 Erik Jonsson School of Engineering and Computer Science Award*: for exceptional poster at SPUR Symposium
- *2023 SIMIODE Challenge Using Differential Equations Modeling (SCUDEM) Outstanding Team Award*

Technical Skills

Programming Languages: Python, C++, Verilog, Bash

Libraries: NumPy, Matplotlib, SciPy, H5Py, Pandas, Tensorflow, Sympy

Laboratory Techniques/Equipment: Network Analyzers, FPGA Design, Wire Bonding, Voltage Probes, Circuit Design

Other Technologies: MATLAB, Vivado, Vitis, Quartus, Linux, Git, Github, SLURM, $\LaTeX$